

Kshitij Soni

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Vehicle Dynamics Simulation Technical Lead | Senior Vehicle Dynamics Engineer | Technical Lead (Systems & Simulation) | IPG Carmaker • MATLAB/Simulink • HCV/LCV Systems

Actively seeking senior simulation/technical lead role • Available immediately • Open to Relocation

Vehicle dynamics simulation technical lead with 6+ years across European R&D, OEM production and EV startup environments. Specialist in multi-body dynamics, IPG CarMaker/TruckMaker, and MATLAB/Simulink for LCV/HCV chassis and powertrain systems. European dual-degree (HAN University of Applied Sciences + Czech Technical University in Prague). Validated simulation methodology achieving 92% accuracy vs VECTO industry standard. Targeting senior simulation or technical lead roles in automotive R&D.

SKILLS & INTERESTS

Compliance, Control Systems, Drive Cycle Validation, Kinematics, Model Based Design (MBD), Multi-body Modelling, Ride & Handling, Systems Modelling, VECTO, Vehicle Dynamics

CAD & Simulation Softwares: Ansys, CATIA, Creo, Fusion 360, IPG CarMaker, MATLAB & Simulink, Onshape, Solidworks, Stateflow

Coding Skills: C++, CANoe, Github, Python

WORK EXPERIENCE

Binani Technologies India Pvt. Ltd. Pune, India • 06/2024 - 03/2026 Technical Lead / Senior Manager - Vehicle Engineering (LCV & HCV) • Full-time

Led the product lifecycle for 3.5 Ton and 9 Ton Battery Electric commercial vehicle platforms using Solidworks and Ansys. Owned FRP body, chassis and battery enclosure design across an 8-month development cycle. Coordinated a cross-functional team of 6 engineers across multiple workstreams, ensuring technical excellence in systems design, high-fidelity simulation, mass-manufacturability and rigorous prototype validation.

• SolidWorks • Ansys • Fusion360 •

Mahindra Vehicle Manufacturers Ltd. Pune, India • 06/2018 - 07/2019 Production Officer (Graduate Apprentice Trainee)

Managed assembly throughput for international LHD/RHD vehicle variants, gaining full-vehicle chassis-to-body integration exposure across production cycles, leading a multi-tiered team to streamline assembly cycles and maintain peak operational productivity.

EDUCATION

European Master Of Automotive Engineering Dual-Degree (ING.) (MSC.) in Vehicle Dynamics and Clean Driveline Control Systems

Czech Technical University in Prague (ČVUT), Prague, Czech Republic and HAN University of Applied Sciences, Arnhem, Netherlands •
GPA: 7.0/10 (Dutch 10-point grading scale)

09/2019 -
01/2023

B. Tech. in Automotive Engineering

SRM Institute of Science and Technology, Chennai, India • GPA: 7.0/10

04/2014 - 03/2018

PROJECTS

Thesis : Total ownership-cost estimation of operating current and future high-capacity vehicles in current infrastructure

12/2021 -
03/2023

HAN Automotive Research, Arnhem

Built MATLAB/Simulink predictive energy and ownership-cost tool validated against VECTO at 92% accuracy across real-world HCV drive cycles, for comparative analysis of conventional, EV and Hybrid heavy-duty powertrains. Owned the end-to-end technical lifecycle of the project under expert research supervision.

· MATLAB/Simulink · StateFlow · Google API · GitHub · VECTO ·

Heavy Vehicle Handling

08/2020 - 01/2021

Ter Linden Transportat B.V., Arnhem

Quantified road transport feasibility for 120m wind turbine blades by engineering a MATLAB/Simulink based tractor-dolly kinematic model with Simulink VD integration, sweep-path analysis for ISO cornering and roundabout maneuvers to validate powertrain requirements and transport strategies.

· MATLAB/Simulink · Google API ·

Driver Feedback Evaluation : Vehicle Testing

11/2020

Rijkswaterstaat NL, Arnhem

Executed dynamic warning bump testing for highway utilities to evaluate vehicle-to-infrastructure safety protocols by conducting in-depth suspension data analysis to quantify damping performance and co-relate with driver feedback during high-stress utility crossings.

· LapSim ·

Performance Camshaft Processing : Technology for Automotive Production

11/2019 - 01/2020

Czech Technical University, Prague

End-to-end production planning for a competition-grade camshaft, utilizing CATIA for process visualization and operation sheet development. Optimized low-volume manufacturing workflows to meet the rigorous tolerances required for aftermarket performance components for the Smart Roadster.

· CATIA V5 ·

Multi-link Driven Rear Suspension System Design

09/2019 - 12/2019

Czech Technical University, Prague

· CATIA V5 · MATLAB/Simulink ·

Thesis : Modelling and Evaluation of a Brake Hydraulics Model with ABS and ESP

12/2017 -
06/2018

Blue Binaries Engineering Solutions Pvt. Ltd., Chennai

Co-developed and validated a high-fidelity MATLAB-based brake system model incorporating ABS and ESP control logic. Orchestrated rigorous Software-in-the-Loop (SiL) tests to assist the team with Hardware-in-the-Loop (HiL) testing protocols to ensure system reliability.

· MATLAB/Simulink · IPG CarMaker · dSPACE · National Instruments ·

Vehicle Dynamics Lead - Formula Student

06/2015 - 04/2018

Camber Racing, SRMIST, Chennai

Led the Vehicle Dynamics department through three vehicle iterations, specializing in tire modeling and steering system optimization before transitioning to a technical leadership role. Orchestrated rigorous testing protocols that directly contributed to top-tier competition performance, including multiple podium placements and international participation in FS Germany.

· SolidWorks · Creo Parametric · OptimumT · OptimumLap · OptimumG · Ansys · MATLAB/Simulink ·

Fabrication and Testing of a composite CVT casing

03/2017

SRMIST, Chennai

Engineered and fabricated a high-performance composite CVT casing for two-wheeler applications using glass fiber reinforcement. Conducted rigorous thermal validation and heat resistance testing to ensure structural integrity under peak operational temperatures.

ACCOMPLISHMENTS

5th Place, Formula Bharat, Coimbatore

01/2018

Camber Racing

3rd Place, Formula Bharat, Coimbatore

01/2017

Team Camber Racing

2nd Place, Formula Student India, Delhi

01/2016

Team Camber Racing

1st Place : Prototype Modelling | Spherical wheeled vehicle and Driveline concept

09/2015

SAE INDIA